

NHS Dental Access in England

Healthcare Business Analysis Case Study | 2019/20 – 2024/25

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Document Type
Independent BA Portfolio Case Study

Background
BDS (Dentistry) | MBA Healthcare Management | CSM

Data Source
NHSBSA Dental Statistics England 2024/25

Tools Used
Power BI · Microsoft Excel · draw.io (BPMN) · NHSBSA Open Data

GitHub
github.com/HariPriyaSilpi/healthcare-ba-portfolio

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21.8M

Pre-COVID Adult Patients (2019/20)

18.0M

Current Adult Patients (2024/25)

-15.4%

Recovery Gap vs Pre-COVID

1.49M

South West Access (lowest region)

3.67M

Midlands Access (highest region)

9x gap

ICB Disparity (GM vs Somerset)

Section 1 — Executive Summary

NHS dental access across England has not recovered to pre-pandemic levels despite five years of gradual improvement. This analysis uses NHSBSA open data (2019/20 to 2024/25) to examine the scale, geography, and demographic distribution of this access gap, and to recommend three targeted interventions.

Problem	Adult NHS dental access remains 15.4% below pre-COVID levels. Only 39.8% of adults were seen in the two years to 2025, versus 49.4% pre-pandemic.
Key Finding	A 9x disparity exists between Greater Manchester ICB (1.11M) and Somerset ICB (123K). South West ICBs consistently record the lowest regional access nationally.
Root Cause	Three structural factors: (1) UDA contract model penalising complex care; (2) rural workforce distribution gaps; (3) incomplete ICB-level data reporting masking true access failure.
Recommendation	A three-component NHS Dental Access Recovery Programme: Regional Triage Hubs in South West ICBs, targeted outreach to 18-64 adults, and a standardised ICB reporting framework.
Author Background	Qualified dentist with NHS clinical experience enabling root cause analysis beyond surface data, including UDA contract dynamics and patient access barriers.

Section 2 — Problem Statement

Despite a gradual post-COVID recovery, NHS dental patient access in England remains significantly below pre-pandemic levels as of 2024/25. Regional and demographic disparities are substantial, and data quality gaps across several Integrated Care Boards suggest the true scale of access inequality may be underreported.

2.1 Scope of this Analysis

- Time period: Financial years 2019/20 through 2024/25 (six years)
- Geography: England — national trend, NHS regional breakdown, and ICB-level comparison
- Population: Adult patients (18+) and child patients (0-17) with age-group sub-analysis
- Metrics: Patients seen, Courses of Treatment (CoT), Units of Dental Activity (UDA)
- Exclusions: Orthodontic-only and domiciliary-only contracts excluded from primary analysis

2.2 Why This Problem Matters

Dental health is directly linked to systemic health outcomes. Untreated dental disease increases risk of cardiovascular disease, diabetes complications, and adverse pregnancy outcomes. The BDA estimates unmet adult dental need has risen from approximately 4 million people in 2019 to 14 million in 2025 — a 250% increase in five years.

Section 3 — Data Analysis and Key Findings

3.1 National Trend — Patients Seen 2019/20 to 2024/25

Year	Adult Patients	Child Patients	Total CoT	Adult % Pop.	Note
2019/20	21,840,358	7,014,122	38,400,000	49.4%	Baseline
2020/21	14,900,000	4,200,000	17,900,000	33.5%	COVID trough
2021/22	16,000,000	5,800,000	26,800,000	35.9%	
2022/23	17,200,000	6,300,000	32,500,000	38.7%	
2023/24	18,000,000	6,700,000	34,000,000	39.7%	Latest
2024/25	18,472,487	6,900,000	35,400,000	39.8%	

Finding 3.1: Adult patients seen in 2024/25 remain 15.4% below the 2019/20 baseline. Total CoT (35.4M) is 10.4% below the pre-pandemic three-year average of 39.6M.

3.2 Regional Analysis — 2024/25

NHS Region	Adult Patients	Child Patients	Access Level
Midlands	3,673,052	1,148,417	HIGHEST
North East and Yorkshire	3,143,321	1,097,973	HIGH
London	2,788,555	993,929	MEDIUM
North West	2,658,830	981,849	MEDIUM
South East	2,627,096	1,077,884	MEDIUM
East of England	2,052,463	832,475	LOW
South West	1,492,030	576,195	LOWEST

Finding 3.2: The South West region records the lowest adult access (1.49M) vs the Midlands (3.67M) — a 2.5x regional gap. At ICB level: Greater Manchester (1,114,602) vs Somerset (122,799) — a 9x difference. This is the most significant finding of this analysis.

3.3 Demographic Analysis

- 18-64 working-age adults: largest group (~13.8M) with greatest absolute deficit — approximately 21% below pre-COVID levels
- 65-74 age group: second largest, showing slower recovery than younger cohorts
- 10-14 children: highest volume child group; orthodontic demand is a key driver
- 75+ age group: smaller volumes but growing unmet domiciliary care demand

3.4 Data Quality Finding

- Several ICBs recorded suppressed or null patient values in the 2024/25 geographical dataset
- Suppressions result from Statistical Disclosure Control (SDC) applied to records with fewer than 5 patients
- True scale of access failure in low-volume ICBs is not fully visible in published data

Finding 3.4: A standardised reporting framework with consistent minimum thresholds is required to enable accurate performance monitoring at ICB level.

Section 4 — Root Cause Analysis

Drawing on clinical experience within the NHS dental system, three structural root causes are identified:

Root Cause 1 — UDA Contract Perverse Incentives

The Unit of Dental Activity (UDA) contract model allocates NHS dental funding based on weighted treatment volumes. The financial penalty for underperforming against UDA targets — clawback of NHS funding — incentivises dentists to focus on high-volume lower-complexity treatments rather than the complex needs of long-waiting patients. This creates a structural disincentive to treat patients with the greatest clinical need.

Root Cause 2 — Rural Workforce Distribution Gap

Dentist density per 100,000 population varies significantly across England by ICB, with South West ICBs recording the lowest ratios. This is compounded by a long-term trend of NHS dentists transitioning to private practice, disproportionately affecting rural areas where private patient demand is lower due to income demographics.

Root Cause 3 — Incomplete ICB-Level Data Reporting

Current NHSBSA reporting applies Statistical Disclosure Control that suppresses small ICB-level values. While this protects patient privacy, it creates blind spots in performance monitoring. NHS England and ICBs cannot accurately identify the worst-performing areas, undermining targeted commissioning and performance accountability.

Section 5 — Recommendation

Based on the findings and root cause analysis, the following three-component NHS Dental Access Recovery Programme is recommended. Each component directly addresses one identified root cause.

#	Component	Description	Root Cause Addressed
1	Regional Dental Triage Hub — South West ICBs	Establish a centralised dental triage hub across the three lowest-access South West ICBs. Patients register via phone or NHS app; a triage coordinator allocates to the nearest NHS practice with available capacity. Target: reduce average wait from 14 weeks to 6 weeks within 18 months.	<i>Root Cause 2: Rural workforce gap. Redirects existing UDA capacity to highest-need patients.</i>
2	Targeted Outreach Programme — Working-Age Adults (18-64)	Commission outreach in South West and East of England ICBs targeting 18-64 adults via employer partnerships and GP referral pathways. Target: 15% increase in 18-64 patients seen within 12 months in priority ICBs.	<i>Root Cause 2: Addresses demand-side access barriers in underserved working-age population.</i>
3	Standardised ICB-Level Reporting Framework	Implement mandatory quarterly reporting for all ICBs with consistent minimum thresholds replacing ad-hoc SDC suppression. Enable real-time performance dashboards for NHS England commissioners. Target: 100% ICB data completeness within 6 months.	<i>Root Cause 3: Eliminates data blind spots and enables performance accountability at commissioning level.</i>

5.1 Expected Benefits

- Reduction in adult access gap from 15.4% to below 10% within 24 months of full programme implementation
- Estimated 500,000 additional adult patients seen annually in South West region at steady state
- Complete ICB-level data visibility enabling targeted commissioning for the first time
- Reduction in preventable dental disease and associated NHS secondary care costs

5.2 Risks and Assumptions

Risk	Likelihood	Mitigation
Insufficient NHS dental workforce in South West to staff Triage Hub	High	Supplement with locum and salaried dental service capacity; explore international recruitment
NHS England funding constraints limit programme rollout	Medium	Phase implementation — pilot one ICB before scaling nationally
ICB resistance to standardised reporting requirements	Low	Mandate via NHS Standard Contract rather than voluntary adoption
Low patient awareness of new triage pathway	Medium	Multi-channel campaign via NHS app, GP practices, and employer networks

Section 6 — AS-IS Process Map: Current State

Figure 1 maps the current-state NHS patient dental access process using BPMN swimlane notation. Four actors are represented: Patient, Dental Practice, NHS England/ICB, and NHS BSA. Pain points identified from the data analysis are annotated at relevant steps.

Figure 1 — AS-IS Process: NHS Patient Dental Access Journey (Current State)

ACTOR / LANE	PROCESS STEP	PAIN POINT / NOTE
PATIENT	START: Patient experiences dental pain or requires routine check-up	
PATIENT	Searches for NHS dentist accepting new patients via NHS website or phone	
DENTAL PRACTICE	[DECISION] Is the practice accepting NHS patients?	YES → Book appointment NO → Patient contacts next practice
<i>⚠ Pain Point 1: 9x variation between highest (GM: 1.11M) and lowest (Somerset: 123K) ICBs. Many South West practices are closed to new NHS patients due to UDA allocation exhaustion.</i>		
PATIENT	Contacts multiple practices — no centralised booking system exists	<i>Patients report calling 10-20 practices before finding NHS availability</i>
PATIENT	[DECISION] Did any practice have NHS availability?	YES → Appointment booked NO → Patient faces unmet dental need
PATIENT	END PATH A: Patient goes private (financial barrier) or waits without treatment	
<i>⚠ Pain Point 2: 14 million adults with unmet dental need (BDA 2025). Low-income and rural patients disproportionately affected.</i>		
DENTAL PRACTICE	Patient attends — practice assesses treatment need and delivers care	
DENTAL PRACTICE	[DECISION] Is UDA allocation for the year remaining?	YES → Treat patient and claim UDA units NO → Redirect to another practice or offer private
<i>⚠ Pain Point 3: UDA clawback mechanism penalises practices for underperforming against targets, creating a structural disincentive against complex treatment delivery.</i>		
NHS ENGLAND / ICB	Commissions NHS dental contracts by region — sets annual UDA targets per practice	
NHS ENGLAND / ICB	[DECISION] Are UDA targets met by the practice?	YES → Full NHS payment released NO → Clawback of NHS funding applied
NHS BSA	Collects treatment data from practices quarterly	

NHS BSA	[DECISION] Is ICB-level data complete and above SDC suppression threshold?	YES → Full regional reporting published NO → Data suppressed — null values recorded in dataset
<i>⚠ Pain Point 4: SDC suppression creates data blind spots at ICB level. Several ICBs recorded null patient values in 2024/25 — true scale of access failure is masked.</i>		
NHS BSA	END: Annual statistics published — access gaps identified but not systematically acted upon at ICB commissioning level	

Section 7 — TO-BE Process Map: Recommended Future State

Figure 2 maps the recommended future-state process following implementation of the three-component NHS Dental Access Recovery Programme. New and improved steps are highlighted in green. Each of the four AS-IS pain points is directly addressed.

Figure 2 — TO-BE Process: NHS Patient Dental Access Journey (Recommended Future State)

ACTOR / LANE	PROCESS STEP	PAIN POINT / NOTE
PATIENT	START: Patient experiences dental pain or requires routine check-up	
PATIENT	NEW: Contacts Regional Dental Triage Hub — single point of contact via phone or NHS app	<i>Addresses Pain Point 1: eliminates need to contact multiple practices</i>
NHS ENGLAND / ICB	NEW: Triage Hub assesses urgency and allocates patient to nearest available NHS practice with UDA capacity	<i>Centralised demand management — matches patient need to available supply</i>
DENTAL PRACTICE	Receives pre-triaged patient referral with clinical urgency level and practice suitability match	<i>Practices receive patients matched to their remaining UDA capacity</i>
NHS ENGLAND / ICB	NEW: Reformed UDA contract: rewards clinical complexity weighting — clawback penalty for complex care removed	<i>Addresses Pain Point 3: removes structural disincentive against treating high-need patients</i>
DENTAL PRACTICE	Treats patient — claims UDA units under reformed contract model	
NHS ENGLAND / ICB	NEW: Targeted outreach programme active in South West and East of England ICBs — GP referral pathways and employer partnerships targeting 18-64 adults	<i>Addresses Pain Point 2: active demand generation in underserved working-age populations</i>
PATIENT	END PATH A (Resolved): Patient receives NHS dental treatment within 6-week target wait	<i>Target: reduce average wait from 14 weeks to 6 weeks within 18 months</i>
NHS BSA	NEW: Standardised ICB reporting framework — mandatory quarterly returns, consistent thresholds, no SDC suppression of operational performance data	<i>Addresses Pain Point 4: full data visibility at ICB level for the first time</i>
NHS ENGLAND / ICB	NEW: Real-time performance dashboard — commissioners identify and act on underperforming ICBs immediately rather than waiting for annual publication	<i>Enables proactive commissioning decisions based on live data</i>
NHS BSA	END: Quarterly data published with complete ICB visibility — enables targeted, evidence-based commissioning intervention	

AS-IS Pain Points Resolved in TO-BE	Key Performance Indicators (TO-BE)
Pain Point 1 resolved: Triage Hub eliminates multi-practice contact barrier	Average dental wait: 14 weeks → 6 weeks (18-month target)

- Pain Point 2 resolved: Outreach programme actively reaches 18-64 adults
- Pain Point 3 resolved: UDA reform removes complex care disincentive
- Pain Point 4 resolved: Standardised reporting eliminates data blind spots

- South West adult access: 1.49M → 1.99M (+500K, 24-month target)
- Adult access gap: -15.4% → below -10% (24-month target)
- ICB data completeness: ~70% → 100% (6-month target)

Section 8 — Author's Note

This case study was produced as an independent business analysis portfolio project to demonstrate end-to-end BA capability: problem definition, data analysis, process mapping, and business case development. All data is publicly available from the NHSBSA and NHS Digital.

The clinical interpretation in Sections 4 and 6 draws on direct experience as a qualified dentist within the NHS dental system, providing contextual depth beyond what is available from data alone. Understanding how UDA contracts affect clinical decision-making, why rural practices struggle to maintain NHS provision, and what the patient access journey looks like from inside a practice — this is the distinctive contribution a Healthcare Business Analyst with clinical experience brings to a pharma or health tech organisation.

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Portfolio

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